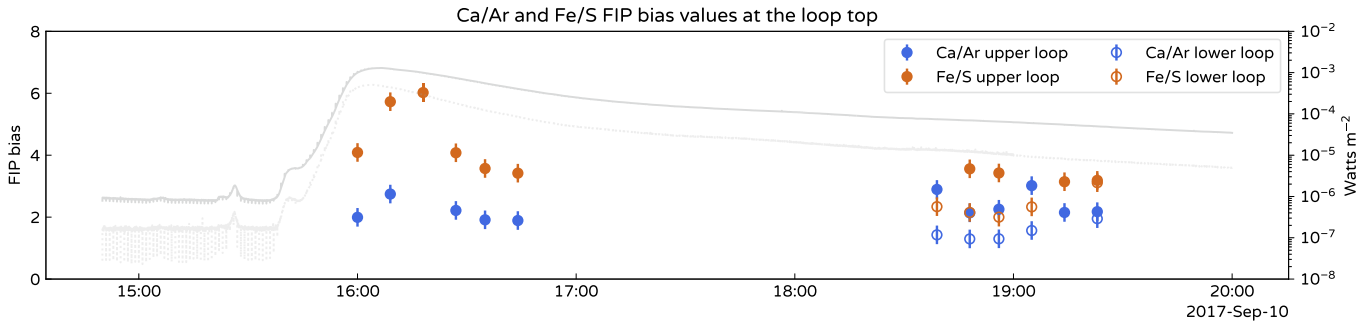

Fig. 2. Ca/Ar and Fe/S FIP bias over time. The blue and orange dots indicate FIP bias calculated using the Ca XIV 193.87 Å/Ar XIV 194.40 Å and Fe XVI 262.98 Å/S XIII 256.69 Å diagnostics, respectively. Vertical bar at each data point indicates an uncertainty of 0.3. Only data points with sufficient χ^2 and good fits are included. The solid and hollow styles indicate composition values calculated using the upper and lower loops, respectively. The lower set of loops can only be identified in the later six rasters. Grey line indicates the GOES X-ray lightcurve of the flare. It is worth noting that recent temporally resolved Sun-as-a-star observations commonly show a decline in FIP bias during flare peaks, contrasting the trend observed in this figure.

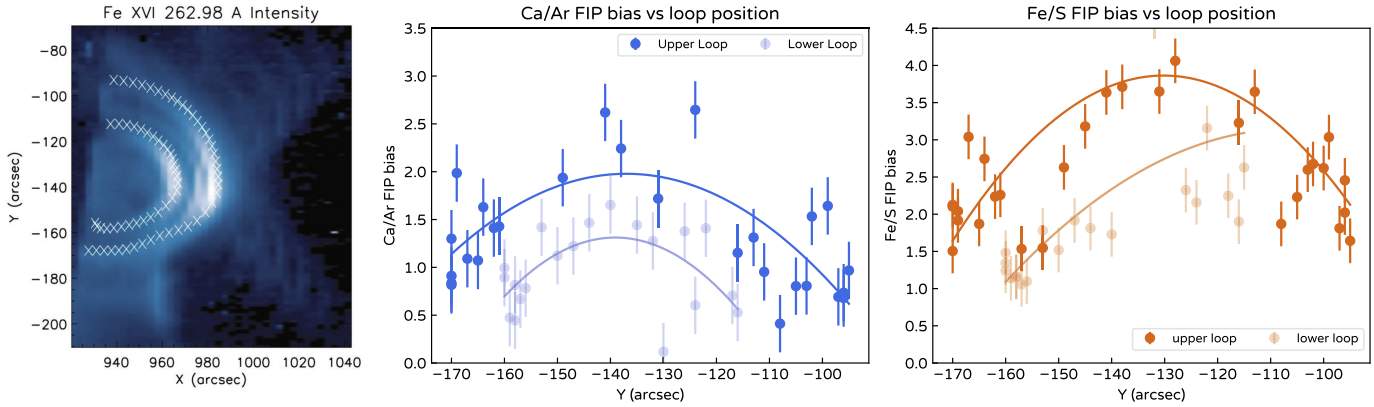

Fig. 3. Ca/Ar and Fe/S FIP bias calculated at positions along the loop marked in the Fe XVI 262.98 Å map at 18:39 UT. Vertical bar at each data point indicates an error of 0.3. Data points with insufficient χ^2 or poor fit are omitted. A correlation between composition and loop position is observed. The loop top exhibits high FIP bias, indicative of fractionated plasma. In contrast, the footpoints show photospheric abundances with the Fe/S diagnostic, and even inverse-FIP abundance with the Ca/Ar diagnostic.

(Ca/Ar). For Ca/Ar, the loop top FIP bias is ~2–3 throughout the flare with the lower loop showing values of ~1.5–2 in the late phase. While this suggests a tendency for the lower loop to have slightly lower FIP bias, the difference is marginal for the Ca/Ar FIP bias diagnostic, and may be close to being within the uncertainties.

3.2. Spatial variation of FIP bias along loop

Figure 3 further examines the spatial variation of composition at 18:39 UT, from loop top to footpoints. Using the same methodology, we calculated the FIP bias along the flare loops. A clear relationship between loop location and FIP bias can be seen in the Ca/Ar vs loop location plot for both the upper and lower flare loops - loop tops exhibit a coronal abundance (FIP bias > 2), gradually decreasing to photospheric abundance (FIP bias = 1) and even potentially inverse-FIP (IFIP) bias (FIP bias < 1, suggesting a depletion of Ca or enhancement of Ar abundance) at the lower loop footpoints. The Fe/S diagnostic shows the same trend at the upper loop, but a weaker trend at the lower loop. Overall, the loop top exhibits coronal abundances, as evidenced by both the Ca/Ar and Fe/S composition diagnostics over time, and loop footpoints are associated with photospheric abundances.

4. Discussion

In this study, we show that composition during flares can be complex, exhibiting significant spatial variation over time. Over 3.5 hours of observations, the flaring loop top consistently shows plasma with high FIP biases in both Ca/Ar and Fe/S diagnostics (FIP bias > 2). FIP bias was higher at the flare peak, decreasing to a slightly milder level in later flare stages. Along the loops themselves, the observed FIP biases decrease toward photospheric (FIP bias = 1) or lower (FIP bias < 1) values as we near the footpoints, in agreement with Doschek et al. (2018).

We propose that this high FIP bias in the loop top indicated by both the Ca/Ar and Fe/S composition diagnostics is due to plasma downflow from the current sheet (CS), or plasma sheet in which it is embedded, and was shown by Warren et al. (2018) to have coronal composition. Figure 4 shows a cartoon illustrating this plasma downflow scenario that we propose, which can contribute to the high FIP bias as quantified by both the Ca/Ar and Fe/S diagnostics we see at the loop top. In past studies, Chen et al. (2015) investigated the AIA 94 Å difference images of a C1.9 class long-duration flare on 2012 March 3. The C-class flare has a similar morphology compared to this X8.2 flare. By investigating the difference images from AIA observations, Chen et al. (2015) show that plasma flows down along the CS recurringly at ~550 km s^{-1}, and this rapidly moving plasma stops